Team 1: Physical Prototype & Hardware / IoT (Agro-Mechanical & Embedded)

1. Enclosure & Deployability (Mechanical)

- **Chamber Design & Fabrication:** Design and 3D print/assemble a compact, deployable chamber (origami fold, telescoping, or modular pop-up mechanism).
- **Environmental Isolation & Seal:** Ensure chamber can maintain light, humidity, and micro-climate control.
- **Growth Medium / Substrate Selection:** Choose a space-grade delivery setup (e.g., aeroponics, hydroponics with rockwool/sponge, or capillary wick systems suited for microgravity simulation).

2. Sensor Integration (IoT Monitoring)

- **Environmental Sensing:** Wire and calibrate sensors for:
    - Ambient temperature & relative humidity ($DHT22$ / $BME280$/ $SHT3x$).
    - Substrate moisture / root water potential.
    - Ambient light intensity / spectrum ($TSL2591$ or photoresistors).
    - $CO_2$and air quality ($SCD30$ or $SGP30$).
- **Visual Monitoring:** Mount a micro-camera (e.g., ESP32-CAM or Raspberry Pi Camera) for periodic visual plant state checks.

3. Actuation & Environmental Control

- **Photoperiod & Spectrum Control:** Build an adjustable LED grow light array (Full Spectrum: 450nm Blue + 660nm Deep Red + Far-Red) with PWM dimming control.
- **Irrigation & Nutrient Delivery:** Implement a peristaltic micro-pump or ultrasonic mister driven by relay/MOSFET drivers.
- **Airflow & Ventilation:** Add micro-fans for boundary layer reduction and humidity circulation.

4. Edge Controller & Telemetry

- **Microcontroller Setup:** Program an ESP32 or Raspberry Pi to read all sensors, control actuators, and expose an API/MQTT broker.
- **Data Pipeline to LLM:** Establish bidirectional communication (MQTT/REST/WebSockets) between the IoT controller and the LLM agent.

Team 2: LLM & Biological Intelligence (AI & Software)

1. Plant Knowledge Base & Context Engine

- **Botanical Knowledge Retrieval (RAG / Structured Database):** Compile species-specific data cards (e.g., Dwarf Tomato, Radish, Microgreens, Lettuce):
    - Optimal VPD(Vapor Pressure Deficit), DLI (Daily Light Integral), and photoperiod.

    - Water cycles, root oxygenation requirements, growth stages (germination →vegetative →flowering →fruiting).

- **Space Stress Factor Rules:** Define baseline constraints for microgravity, cosmic radiation, and closed atmosphere risks (ethylene buildup, high CO_2).

2. LLM Orchestration & Decision-Making Agent

- **Sensor Telemetry Parsing:** Build structured input schemas (JSON) to pass real-time sensor readings + current crop stage to the LLM.
- **Dynamic Condition Adaptation:** Prompt-engineer/fine-tune the LLM to output actionable control vectors:
    - *Example output:* {"target_irrigation_ml": 45, "light_spectrum_ratio_r_b": 4.5, "light_intensity_pct": 80, "ventilation_fan_duty": 60, "reasoning": "Early fruiting stage detected; increasing red-spectrum DLI and moderating root saturation."}
- **Anomaly Detection & Diagnosis:** Enable the LLM to detect anomalous sensor trends (e.g., sudden moisture drop, sensor drift, stagnant transpiration) and produce troubleshooting directives.

3. Safety Guardrails & Validation Layer

- **Failsafe Envelope Filter:** Implement deterministic hard safety bounds (e.g., never allow LED temp $> 60°$C, never run pump dry, never drop moisture below critical wilting point) to validate LLM outputs before sending instructions to the physical actuators.
- **Astronaut UI / Chat Interface:** Develop a lightweight dashboard or conversational terminal where astronauts can ask: *"What is the status of the lettuce?"* or *"When will tomatoes be ready to harvest?"*

Team 3: Poster Design & Visual Communication

1. Narrative Architecture & Visual Flow

- **Layout Grid & Hierarchy (Standard 3-Column / Modular Flow):**
    1. **Hook & Problem Statement:** Deep space nutrition challenges, psychology of fresh food, zero-G agricultural constraints.
    2. **Core Architecture:** Dual-engine loop (Physical Deployable Phytotron + Cognitive LLM Feedback Loop).
    3. **Data & Results:** Prototype sensor telemetries, LLM decision curves, space-resource efficiency metrics.
    4. **Impact & Future Horizons:** Mars missions, Artemis base viability, Earth-side spin-offs (desert farming, urban ag).

2. Diagram & Asset Creation

- **Hero System Schematic:** Render/illustrate the deployable greenhouse cutaway showing:
    - Folding frame / deployable geometry.

    - LED spectrum arrays, hydroponic/aeroponic capillary root zone.
    - Sensor array placement.
- **The "Sense →Think →Act" Pipeline Flowchart:**
    - Clean, polished flowchart showing: Sensors (IoT) -> Cloud/Local LLM Brain -> Actuators (Pumps/Lights) -> Plant Response.
- **LLM Decision Showcase:** Create a clean graphic mockup comparing standard static farming vs. LLM dynamic adaptation during an environmental disturbance.

3. Poster Execution & Polish

- **Typography & Color Palette:** Use an aerospace/futuristic palette (Deep space navy/carbon #0B0F19, bio-green #10B981, cyber cyan #06B6D4, crisp white text).
- **QR Code & Interactive Demo:** Generate a QR code linking to a GitHub repository, live telemetry dashboard, or video demonstration of the physical deployable prototype.
- **Print Preparation:** Ensure 300 DPI, correct bleed margins, CMYK color proofing, and readable font sizes from 1.5–2 meters away.